

MSc Thesis Defense Committee

Evaluation Report

Committee Simulation for MSc Thesis Defense

Candidate: Arlen Smagulov

Thesis: Pose Guided Predictive Ballistics for Body Part-Targeted Football Training

Supervisor: Prof. Sultangali Arzykulov | Co-Supervisor: Prof. Mohammad Hashmi

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Committee Roles (per NU MSc ECE Handbook):

Lead Supervisor (S1): 35% weight

Co-Supervisor (S2): 35% weight

External Examiner (EE): 30% weight

Grading: $FE = 0.7 \cdot M + 0.3 \cdot O$, where O = oral, M = manuscript

PASS threshold: $FE \geq 75\%$

This document evaluates the ORAL DEFENSE only (O component).

Scale: 1-10 per criterion, mapped to handbook criteria.

SECTION 1: PER-SLIDE EVALUATION

Scoring: 1-10 scale per criterion per committee member.

S1 = Lead Supervisor (35%), S2 = Co-Supervisor (35%), EE = External Examiner (30%)

SLIDE 1: Title Slide

Speech_Style: S1=8 S2=8 EE=7 -- Clean professional title; no speech to evaluate yet. Handbook: 'Professional appearance' met

Structure: S1=8 S2=7 EE=8 -- Logical opening; thesis title matches manuscript exactly. Handbook: 'Logical sequence' starts well

Layout: S1=8 S2=8 EE=8 -- Uncluttered; two logos, name, supervisors. Handbook: 'Uncluttered' satisfied.

Flags:

No flags. Clean opening slide.

SLIDE 2: Motivation

Speech_Style: S1=8 S2=7 EE=8 -- Bullets are concise and use correct terms. Handbook: 'Correct use of terms' met.

Structure: S1=8 S2=8 EE=7 -- Logical problem buildup; no literature reference cited on slide. Handbook: 'Supporting body of literature' met

Layout: S1=7 S2=7 EE=7 -- Two images take substantial space; 5 bullets + 2 figures + takeaway = borderline cluttered. Handbook: 'Cluttered' not met

Intro_Lit: S1=7 S2=7 EE=6 -- Problem statement is clear; but no literature cited (OptiTrack/Vicon mentioned in images only). Handbook: 'Literature' not met

Tech_Competency: S1=7 S2=7 EE=7 -- Gap identified but justification of approach deferred. Handbook: 'Justification of approach' not met

Flags:

MISSING LITERATURE: No citation or author/year on slide. Handbook requires 'Supporting body of literature mentioned' (Structure 10)

COST CLAIM: '~USD 200 perception hardware' is accurate per thesis p.3, but total BLM cost (~USD 358) not clarified until Slide 8.

NU-Compliant Rewrite (max 6 bullets):

Motivation: From Open-Loop Launchers to Pose-Reactive Training

- Commercial launchers (Lobster Elite, Spinshot) fire fixed trajectories -- no athlete awareness [1,2]
- Lab MoCap (OptiTrack \$50k-200k, Vicon) achieves sub-mm accuracy but requires markers and has no actuation [1,2]
- Gap: No system combines low-cost markerless 3D perception with physical launcher actuation
- This thesis: 4 commodity USB cameras (\$120 total) + YOLO-Pose + DLT/SVD triangulation -> pose-reactive aiming
- Validation boundary: closed-loop firing at a moving subject is NOT demonstrated (Sec 1.3, p.4)

SLIDE 3: Problem Statement & Objectives

Speech_Style: S1=8 S2=8 EE=8 -- Precise numerical targets and achievements. Handbook: 'Correct use of terms' and 'clear' satisfied

Structure: S1=9 S2=9 EE=8 -- Excellent RQ-based structure mirroring thesis Section 1.2. Handbook: 'Key points sufficiently high-level' met

Layout: S1=7 S2=7 EE=6 -- Four objective boxes + takeaway + all numbers = dense. Handbook: 'Concise & precise' partially met

Intro_Lit: S1=8 S2=8 EE=8 -- Clear problem statement with measurable objectives. Handbook: 'Problem statement & project objectives' met

Tech_Competency: S1=8 S2=8 EE=7 -- Quantified objectives good; but 'all four objectives are met' claim needs nuance -- RQ3 satisfied

Flags:

OVERSTATING: Takeaway says 'all four objectives are met' but RQ3 is satisfied only for static/live-aim, not moving-target (p.4, Sec 1.3)

143.38mm figure is from 62 valid trials out of 81 (76.5% validity). The 19 invalid trials are a limitation not mentioned here (Sec 5.4, p.49)

NU-Compliant Rewrite (max 6 bullets):

Problem Statement & Research Objectives

- RQ1: Ball localisation mean 3D error < 120mm -> Achieved: 95.17mm (36/36 trials, Table 5.1)
- RQ2: Joint localisation mean 3D error < 180mm -> Achieved: 143.38mm (62/81 valid, 76.5% validity, Table 5.2)
- RQ3: Safety-gated integration -> S0-S4 passed 2026-04-09; moving-target closed-loop NOT validated (Sec 1.3)
- RQ4: Commodity hardware, ~USD 200 perception, open-source stack -> Wired and dry-run validated
- NOTE: RQ3/RQ4 are satisfied within static/live-aim scope only; moving-target firing is future work (Sec 6.4)

SLIDE 4: Background & Research Gap

Speech_Style: S1=8 S2=8 EE=8 -- Tabular comparison is professional. Handbook: 'Easy-to-understand sequence' met.
Structure: S1=9 S2=9 EE=9 -- Excellent gap analysis with structured comparison. Handbook: 'Supporting body of literature mentioned' met.
Layout: S1=9 S2=9 EE=8 -- Good use of table; concise text. Handbook: 'Good use of charts, tables' satisfied.
Intro_Lit: S1=8 S2=8 EE=8 -- Literature categories well covered. Handbook: 'Literature review and conclusions' addressed through intro.
Tech_Competency: S1=7 S2=7 EE=7 -- Gap identified clearly; but 'Static/live-aim validated' in 'This work' column could imply more.

Flags:

MISLEADING LABEL: Table column 'Closed-loop' says 'Static/live-aim validated' for This Work. The thesis explicitly states closed-loop vision-guided sports robots row says 'Fixed zones' -- this needs a citation (e.g., Footbot [47] mentioned in Sec 6.4.7).

NU-Compliant Rewrite (max 6 bullets):

- Background & Research Gap
- Optical MoCap (OptiTrack, Vicon): \$50k-200k, sub-mm, marker-based, no launcher [1,2]
- Commercial launchers (Lobster Elite, Spinshot): \$200-4k, open-loop only, no perception [Sec 2.6]
- Vision-guided sports robots: \$2k-10k, fixed-zone targeting, no joint-level reactivity
- This work: ~\$200 perception, targeting-level accuracy, markerless, aim + controlled single-shot validated
- Key difference: pose-reactive aiming from live 3D joints, not pre-programmed zones
- UNVALIDATED: closed-loop firing at a moving subject (Sec 1.3, p.4)

SLIDE 5: Key Technical Components (6 Contributions)

Speech_Style: S1=8 S2=8 EE=7 -- Good terminology; '6.2 ms/frame TRT FP16' is precise. Handbook: 'Correct use of terms' met.
Structure: S1=8 S2=8 EE=8 -- Six numbered contributions mirror thesis Section 1.4. Handbook: 'Development of topics described' met.
Layout: S1=7 S2=7 EE=6 -- Six boxes + takeaway = dense. Handbook: 'Uncluttered' borderline.
Intro_Lit: S1=7 S2=7 EE=7 -- Contributions listed but not contextualized against literature. Handbook: 'Relevance to industry/society' met.
Tech_Competency: S1=8 S2=8 EE=7 -- Good breadth; '95.17 mm corrected ball mean' uses thesis data correctly. But contribution 03 needs more context.

Flags:

OVERSTATING AUTONOMY: Contribution 01 says 'Markerless, closed-loop, commodity hardware'. Thesis p.4 states closed-loop at motion capture. Contribution 03: '15 FPS live pipeline' is correct per Sec 3.3, but 6.2ms is per-batch inference, not total pipeline time (~15ms per Sec 3.3).

NU-Compliant Rewrite (max 6 bullets):

- Key Technical Contributions
- C1: Pose-reactive aiming from live 3D joint reconstruction (aim + controlled single-shot, Sec 5.9)
- C2: 4-camera DLT/SVD targeting pipeline, 95.17mm corrected ball mean (Table 5.1)
- C3: YOLO-Pose TRT FP16 at 6.2ms/batch within 15 FPS pipeline (~15ms total, Sec 3.3)
- C4: Six-stage safety protocol S0-S4 passed, E-STOP <100ms (Sec 3.14, Appendix A)
- C5: Vosk offline ASR + 16-phrase grammar behind safety gates (Sec 3.12)
- C6: 36-pt ball grid + 81-trial joint-touch protocol with JSONL decision logs (Ch 4)

SLIDE 6: Proposed System - Pose-to-Aim Integration

Speech_Style: S1=7 S2=7 EE=7 -- Figure-dependent slide; verbal explanation needed. Handbook: 'Professional delivery' requires more.
Structure: S1=8 S2=8 EE=8 -- Logical system-level view before diving into details. Handbook: 'Smooth progression from topic to topic' met.
Layout: S1=8 S2=8 EE=8 -- Single figure + takeaway = clean. Handbook: 'Uncluttered' satisfied.
Tech_Competency: S1=8 S2=8 EE=8 -- Integration concept clearly shown. Handbook: 'Viability of design' supported.

Flags:

No major flags. Good architectural overview.

SLIDE 7: Architecture & Pipeline

Speech_Style: S1=8 S2=8 EE=8 -- Precise latency breakdown. Handbook: 'Correct use of terms' and 'clear' met.
Structure: S1=9 S2=9 EE=9 -- Excellent end-to-end flow mirroring Sec 3.3. Handbook: 'Logical sequence, good flow'.
Layout: S1=8 S2=8 EE=8 -- Pipeline diagram + timing = informative and uncluttered. Handbook: 'Good use of diagrams'.
Tech_Competency: S1=9 S2=9 EE=8 -- Strong justification: 52ms headroom quantified. Handbook: 'Justification of approach' and

Flags:

No flags. Excellent technical slide.

SLIDE 8: Hardware Implementation

Speech_Style: S1=8 S2=8 EE=8 -- Cost table is a strong addition. Handbook: 'Professional delivery' with concrete numbers.
Structure: S1=8 S2=8 EE=8 -- Hardware -> cost -> practical details flows well. Handbook: 'Development of topics clearly'.
Layout: S1=7 S2=7 EE=6 -- 6 images + cost table + 2 text blocks + takeaway = cluttered. Handbook: 'Uncluttered' violated.
Tech_Competency: S1=9 S2=9 EE=8 -- Cost analysis directly addresses handbook: 'Costs and efficiency considerations'. Total -

Flags:

CLUTTER: 6 hardware photos compete with cost table. Reduce to 2-3 key images max.

Cost table total is ~\$358 (not \$200). The \$200 figure is perception-only. Be explicit: 'Perception ~\$120; total BLM ~\$358'.

NU-Compliant Rewrite (max 6 bullets):

- Hardware Implementation (~USD 358 total)
- 4x Hikvision DS-E12 USB cameras (1280x720, ~\$30 each) at arena corners (Sec 3.2.1)
- ESP32 MCU: cooperative FSM at 921,600 baud, dual-core, ~\$8 (Sec 3.2.3)
- NEMA-23 steppers + worm-gear reducers: self-locking for safety (Sec 3.2.2)
- Perception PC (recycled): handles YOLO + Kalman + solver (Sec 3.3)
- Cost: Perception ~\$120 | Actuators ~\$90 | Flywheels ~\$60 | Chassis ~\$80 | Total ~\$358
- AprilTag fiducials on walls for extrinsic calibration (Sec 3.5)

SLIDE 9: Software Implementation

Speech_Style: S1=8 S2=8 EE=8 -- Layered architecture table is clear. Handbook: 'Correct use of terms' met.
Structure: S1=8 S2=8 EE=8 -- Software stack flows top-to-bottom. Handbook: 'Logical sequence'.
Layout: S1=7 S2=7 EE=7 -- Table + code excerpt + takeaway = moderate density. Code excerpt is useful but adds clutter. Handbook: 'Uncluttered'.
Tech_Competency: S1=9 S2=9 EE=8 -- Shows tools/software used (handbook criterion); TensorRT FP16, SVD-DLT, CV Kalman

Flags:

Code excerpt shows `handle_set()` -- good for demonstrating firmware gates. Consider moving to Appendix to reduce clutter.

Missing: 'iterative reprojection-error rejection' for ball (Sec 3.7.2) and 'adaptive EMA' should be mentioned in Filter row.

SLIDE 10: Calibration & Geometry

Speech_Style: S1=7 S2=7 EE=7 -- Figure-heavy; verbal explanation critical. Handbook: 'Clear and easily understood' depends on
Structure: S1=8 S2=8 EE=8 -- Calibration is correctly sequenced before triangulation. Handbook: 'Logical sequence'.
Layout: S1=8 S2=8 EE=8 -- Single figure + takeaway = clean. Handbook: 'Uncluttered'.
Tech_Competency: S1=8 S2=8 EE=8 -- Calibration is critical for 3D accuracy. Handbook: 'Justification of approach' -- ChArUco +

Flags:

SLIDE LACKS NUMBERS: Intrinsic reprojection error 2-8 px (Sec 5.1); extrinsic RMSE 3-7 px (Sec 5.2). These should be on the slide.

Missing: mention of PnP with RANSAC + sigma-clipping for robustness (Sec 3.5.2).

NU-Compliant Rewrite (max 6 bullets):

- Calibration & Geometry
- Intrinsic: ChArUco board (5x7 squares, DICT_4X4_1000), Zhang method [18], reprojection 2-8 px (Sec 5.1)

- Extrinsic: 24 AprilTag fiducials on walls, robust PnP + RANSAC + sigma-clipping (Sec 3.5.2)
- Extrinsic RMSE: 3-7 px after outlier rejection; 8-15% of tag corners removed (Sec 5.2)
- Overlay validation: reprojected corners within 5-10 px of detected across all 4 cameras (Sec 5.2)
- Result: unified millimetre-scale world frame enabling 3D joint coordinates from 4 USB cameras

SLIDE 11: Methodology & Experiments

Speech_Style: S1=8 S2=8 EE=8 -- Tabular results summary is professional. Handbook: 'Clear and easily understood'.

Structure: S1=9 S2=9 EE=9 -- Excellent structured overview of all experiments. Handbook: 'Key points sufficiently highlighted'.

Layout: S1=8 S2=8 EE=8 -- Clean table format. Handbook: 'Good use of tables'.

Tech_Competency: S1=8 S2=8 EE=7 -- Good breadth; but '3D trajectory verified' for dynamic clips is vague. Handbook: 'Test pro

Flags:

VAGUE: '3D trajectory verified' for dynamic clips -- state the actual pass criteria from Sec 4.3: <800mm frame-to-frame jumps for slow,

Bias correction row says 'Ball mean drops to 95.17mm' -- raw was 150.77mm (Fig 5.1). Show both numbers.

NU-Compliant Rewrite (max 6 bullets):

Methodology & Experiments

- Ball GT: 36-point 3D grid (3x3x4), hold-window stats -> mean 95.17mm corrected (raw 150.77mm, Table 5.1)
- Joint-touch: 81 trials, 62 valid (76.5%), per-joint: knee 110 / hip 150 / shoulder 164mm (Table 5.3)
- Dynamic clips: ball_slow (<800mm jumps), ball_fast (motion blur stress), no_ball (near-zero FP, Sec 4.3)
- Bias correction: linear per-axis scale+offset, raw 150.77mm -> corrected 95.17mm (Fig 5.1, Sec 4.4.2)
- Safety: S0-S4 passed 2026-04-09, E-STOP <100ms, all JSONL logged (Appendix A)
- NOTE: bias correction fitted in-sample on same 36-point dataset (Sec 4.4.2, limitation in Sec 6.3)

SLIDE 12: Key Results - Accuracy, Safety, Runtime

Speech_Style: S1=8 S2=8 EE=8 -- Headline numbers are clear and precise. Handbook: 'Correct use of terms' met.

Structure: S1=9 S2=9 EE=9 -- Results are the natural climax of the methodology. Handbook: 'Key points sufficiently highlighted'.

Layout: S1=8 S2=8 EE=7 -- Headline boxes + 2 figures + takeaway = moderately dense. Handbook: 'Good use of charts' satisfied

Tech_Competency: S1=9 S2=9 EE=8 -- All key metrics present with exact thesis numbers. Handbook: 'Interpretation of results' s

Flags:

MISSING P95: Ball P95 = 166.51mm, Joint P95 = 198.73mm (Tables 5.1, 5.2). These are important for safety/reliability assessment.

Per-joint errors shown (110/150/164) but P95 values (170.75 / 172.31 / 199.54mm per Table 5.3) are absent.

Raw vs corrected comparison (150.77mm -> 95.17mm) from Fig 5.1 should be explicitly stated.

NU-Compliant Rewrite (max 6 bullets):

Key Results - Accuracy, Safety, Runtime

- Ball: mean 95.17mm, P95 166.51mm (raw 150.77mm -> corrected, Table 5.1)
- Joint: mean 143.38mm, P95 198.73mm (62/81 valid, Table 5.2)
- Per-joint: knee 110.0mm, hip 150.4mm, shoulder 164.4mm (P95: 171/172/200mm, Table 5.3)
- E-STOP: <100ms latch response, tested 3x on 2026-04-09 (Sec 5.9)
- Pipeline: ~15ms total, 52ms headroom within 67ms budget at 15 FPS (Sec 3.3)
- All acceptance thresholds met WITHIN validated scope (static/live-aim only)

SLIDE 13: Limitations & Challenges

Speech_Style: S1=8 S2=8 EE=9 -- Honest limitation framing impresses external examiner. Handbook: 'Correct use of terms' and

Structure: S1=8 S2=8 EE=8 -- Validated/pending split is effective. Handbook: 'Key challenges sufficiently highlighted'.

Layout: S1=8 S2=8 EE=8 -- Figure + clear text = uncluttered. Handbook: 'Uncluttered' satisfied.

Tech_Competency: S1=8 S2=8 EE=8 -- Good: explicitly states unvalidated boundary. But missing key limitations from Sec 6.3: in

Flags:

INCOMPLETE LIMITATIONS: Sec 6.3 lists 8 limitations; slide mentions only 1 (moving-target). Must include: (1) in-sample bias correction. Figure appears to show error distribution -- good, but should be explicitly labeled as Fig 5.4 (joint-touch boxplot).

NU-Compliant Rewrite (max 6 bullets):

Limitations & Challenges

- UNVALIDATED: closed-loop firing at a moving subject (dominant boundary, Sec 1.3, 6.3)
- In-sample bias correction: fitted on same 36-pt dataset, no held-out set (Sec 4.4.2, 6.3)
- 3-camera minimum: 19/81 joint-touch trials invalid (23.5%) due to occlusion (Sec 5.4)
- RPM-to-velocity not empirically calibrated: nose shots at 800 RPM off-target (Sec 5.9)
- Single-person, single-arena, indoor-only evaluation (Sec 1.3, 6.3)
- CV Kalman neutral on jump sequences -- IMM filter needed (Sec 5.7, 6.4)

SLIDE 14: Conclusions & Contributions

Speech_Style: S1=8 S2=8 EE=8 -- Concise restatement of contributions. Handbook: 'Correct use of terms'.

Structure: S1=8 S2=8 EE=8 -- Mirrors Section 6.1 contributions. Handbook: 'Logical sequence'.

Layout: S1=8 S2=8 EE=8 -- Four contribution boxes + takeaway = clean. Handbook: 'Uncluttered'.

Tech_Competency: S1=7 S2=7 EE=7 -- Takeaway says 'strong partial validation' which is honest. But contribution 2 says '~USD

Flags:

COST AMBIGUITY: Contribution 2 says '~USD 200' but total BLM hardware is ~\$358 (Slide 8 table). Clarify.

Missing: 'advantages and disadvantages' per handbook criterion 7 (Conclusions). No disadvantages listed.

NU-Compliant Rewrite (max 6 bullets):

Conclusions & Contributions

- C1: BLM autonomously aims from live 3D joints -- first pose-reactive launcher (Sec 6.1)
- C2: 4 USB cameras, ~\$120 perception, achieve 95.17mm ball / 143.38mm joint accuracy
- C3: Six-stage safety protocol S0-S4 passed, E-STOP <100ms, full JSONL logging (Sec 3.14)
- C4: Voice + keyboard multi-modal control behind safety gates (Sec 3.12)
- Key disadvantage: closed-loop at moving target NOT validated (dominant limitation)
- Other disadvantages: in-sample bias fit, 3-camera occlusion constraint, no RPM calibration

SLIDE 15: Future Work + Ethics & Standards

Speech_Style: S1=8 S2=8 EE=8 -- Standards listed correctly. Handbook: 'Correct use of terms'.

Structure: S1=8 S2=8 EE=7 -- Future work + ethics on one slide is compressed. Handbook: 'Smooth progression' -- ethics deserves

Layout: S1=7 S2=7 EE=6 -- Two columns (Future + Ethics) + takeaway = dense. Handbook: 'Concise' borderline.

Tech_Competency: S1=7 S2=7 EE=7 -- Standards alignment is good but shallow. Handbook: 'Compliance with good practices and

Flags:

MISSING DUAL-USE: Sec 6.5 (p.64) explicitly states: 'A system capable of autonomously tracking human body parts and directing a p

SHALLOW STANDARDS: Listing ISO numbers without explaining HOW they are implemented is insufficient. Add: ISO 13849-1 -> L8

Missing: IEC 62061 from thesis Sec 3.14.2 alignment discussion.

NU-Compliant Rewrite (max 6 bullets):

Future Work & Professional Ethics

- F1: Closed-loop moving-target firing (Stages S5/S6, Sec 6.4.1)
- F2: Empirical RPM-to-velocity calibration via Doppler radar (Sec 6.4.3)
- F3: IMM filter for jump-class motion (Sec 5.7, 6.4)
- F4: Held-out calibration grid for out-of-sample validation (Sec 6.4.6)
- Safety: ISO 13849-1 Cat-1 stop (NC E-STOP, L8), IEC 60204-1 (24V/50A fuse), ISO 10218-1 (exclusion zone)
- Dual-use: system can track body parts and launch projectiles -- deployment requires same gating discipline (Sec 6.5)

SECTION 2: SIMULATED Q&A (Criterion 4, 20 points)

12 questions weighted to methodology, results, safety/ethics, novelty, and future work.

For each: question, your likely answer grading (Pass/Weak/Fail), and model answer with thesis page references.

Q1 (Methodology): Why did you choose four cameras rather than two or three? What happens if

Likely Answer Grade: Pass

Model Answer:

Four cameras provide redundancy: DLT/SVD triangulation requires minimum 2 cameras, but 3+ makes the system over-determined and robust to per-camera noise (Sec 2.1, p.7; Sec 3.6). If one camera is occluded, three remain for triangulation. The minimum-camera-count threshold is configurable (default 2 for ball, 3 for joints). The 4-camera overlap ensures that even at arena edges, most joints are visible to at least 3 cameras (Sec 3.7.5, p.26). Two cameras would provide only a baseline-dependent depth estimate sensitive to calibration error; three is the minimum for reliable joint tracking; four adds the redundancy needed for a safety-gated system (Sec 2.9.2, p.15).

Q2 (Methodology): How does AprilTag-based extrinsic calibration work, and what is the PnP rob

Likely Answer Grade: Pass

Model Answer:

Twenty-four AprilTag fiducial markers are placed on the arena walls at known world-frame coordinates (Sec 3.5.1, p.23). Each tag provides 4 corner points with known 3D positions; the Perspective-n-Point algorithm recovers the camera pose (R, T) by minimising reprojection error. Robustness is achieved through RANSAC-based outlier rejection followed by iteratively re-weighted least squares with sigma-clipping at scale=2.0, which removes 8-15% of tag corner observations, primarily from tags at oblique angles (Sec 3.5.2, p.23; Sec 5.2, p.45). Overlay validation confirms reprojected corners within 5-10 px of detected positions across all cameras (Sec 5.2).

Q3 (Results): The raw ball error was 150.77mm with axis biases of $e_x=+55\text{mm}$, $e_y=+14\text{mm}$, $e_z=-1$

Likely Answer Grade: Weak

Model Answer:

The $e_z = -103\text{mm}$ bias means the triangulated ball positions are systematically placed approximately 103mm below their true elevation (Sec 4.4.2, p.40; Fig 5.1). This is consistent with all four cameras being mounted near the ceiling: at low Z (200-700mm from floor), the shallow downward-looking ray angles are geometrically sensitive to small calibration errors in the extrinsic Z positions of the cameras, causing triangulated points to be placed below their true elevation (Sec 5.3, p.46). The linear per-axis bias correction model (scale + offset per axis) reduces the mean from 150.77mm to 95.17mm (Fig 5.1). However, this correction is fitted in-sample on the same 36-point dataset, so it may underestimate generalisation error (Sec 4.4.2, p.41; Sec 6.3).

Q4 (Results): Why are joint errors 110-164mm, increasing from knee to shoulder? What causes t

Likely Answer Grade: Pass

Model Answer:

Joint error increases monotonically from right_knee (110.03mm) to right_hip (150.38mm) to left_shoulder (164.38mm) because higher joints are closer to the ceiling-mounted cameras and are observed at increasingly oblique angles, weakening triangulation geometry (Sec 5.4, p.49; Fig 5.6). Additionally, the left shoulder is the joint most likely to be occluded by the subject's head and neck. The 19/81 invalid trials (23.5%) occur primarily because the subject's body occluded the target

joint from one or more cameras, reducing the camera count below the minimum-3 threshold for accurate joint triangulation (Sec 5.4, p.49; Sec 4.2.2, p.38). This is a real operational constraint of the system.

Q5 (Safety): Describe your six-stage integration protocol. What would happen at Stage S4 if the

Likely Answer Grade: Pass

Model Answer:

The six stages are: S0 (serial connectivity), S1 (manual ESP32 commands), S2 (runtime without cameras, synthetic targets), S3 (live aim-only, no ball), S4 (safety verification: E-STOP, latch, link-loss), S5/S6 (controlled fire, full-cycle reliability) (Sec 3.14.3, p.35; Appendix A). The protocol is incremental: each stage validates a component before the next is exercised. If the E-STOP failed to latch at S4, the system would NOT proceed to S5 -- the stage has a defined pass criterion requiring E-STOP response <100ms and latch hold until manual clear (Sec 3.14.3). Additionally, the NC (normally-closed) mushroom switch design means a broken wire fails to safe (de-energised motor rail), and the link-loss watchdog on L9 auto-sends stop + set 0 0 0 0 on serial/UDP timeout (Sec 3.14.1, p.35).

Q6 (Safety/Ethics): How does your system align with ISO 10218 and ISO 13849? What is the dual

Likely Answer Grade: Weak

Model Answer:

ISO 13849-1 alignment: the E-STOP at Layer 8 implements a Category 1 stop by removing energy from the motor rail via a normally-closed switch, with two-action release (manual clear + physical reset) (Sec 3.14.2, p.35). ISO 10218-1: exclusion-zone discipline during controlled fire, with the operator behind the launcher and no person down-range unless the operator is the designated target (Sec 3.14.2). ISO 12100: general machinery safety principles applied to the L1-L10 hazard analysis. IEC 60204-1: 24V/50A fuse sizing and single star-point ground topology (Sec 3.2.2, p.19). Dual-use: Section 6.5 (p.64) explicitly states that 'a system capable of autonomously tracking human body parts and directing a projectile at them has obvious dual-use potential beyond sports training.' The author emphasises that the safety architecture (operator presence, hardware E-STOP, explicit target designation, six-stage protocol) is the necessary safeguard for any deployment outside controlled research.

Q7 (Novelty): How does your system differ from OptiTrack or Vicon-based motion capture? Why

Likely Answer Grade: Pass

Model Answer:

OptiTrack and Vicon are observation-only systems: they reconstruct 3D positions to sub-mm accuracy but have no integrated actuation and no launcher control (Sec 1.1, p.1; Sec 2.1, p.7). They cost USD 50,000-200,000, require dedicated studios and reflective markers on subjects, and are inaccessible to community clubs, physiotherapy practices, and individual training (Sec 1.1). This thesis demonstrates that commodity USB cameras at approximately USD 30 each, with markerless YOLO-Pose estimation, can achieve targeting-level accuracy (95.17mm ball, 143.38mm joint) in a domestic garage without markers. The key difference is not accuracy but the combination of low-cost markerless perception WITH physical launcher actuation, at 1-3 orders of magnitude lower cost (Sec 2.9.3, p.16).

Q8 (Novelty): How does your system differ from a Lobster Elite ball launcher?

Likely Answer Grade: Pass

Model Answer:

The Lobster Elite is a conventional open-loop launcher: a human sets the angle, speed, and interval, and the machine repeats that programme indefinitely with no awareness of the athlete's position (Sec 1.1, p.1; Table 2.1, p.14). This thesis's

BLM autonomously computes pitch, yaw, and wheel RPM from real-time 3D joint coordinates derived from multi-view pose estimation (Sec 1.4, p.4). The operator selects which joint to target but does not specify the trajectory. The Lobster Elite costs USD 200-4,000; this system's perception hardware costs approximately USD 120, with total hardware approximately USD 358 (Sec 3.2, Table in Slide 8). The qualitative departure is: the launcher decides its own aim based on live observation (Sec 6.1, p.57).

Q9 (Results): Your corrected ball accuracy of 95.17mm relies on in-sample bias correction. How

Likely Answer Grade: Weak

Model Answer:

The bias and scale parameters of the linear correction model were estimated from the same 36-point static ball dataset used for final accuracy reporting (Sec 4.4.2, p.41). This means the corrected figures (95.17mm mean, 166.51mm P95) should be interpreted as a near-upper bound on the dataset rather than out-of-sample generalisation (Sec 5.3, p.46; Sec 6.3, p.59). With a held-out calibration set, the corrected error would likely be somewhat higher, because the in-sample fit can exploit dataset-specific noise patterns. The thesis explicitly identifies this as a limitation (Sec 6.3) and proposes a held-out calibration grid as future work (Sec 6.4.6, p.62). The temporal precision of 3.79mm (P95: 8.51mm) demonstrates that the dominant error source is systematic calibration bias rather than random noise, which is exactly what the correction model addresses (Table 5.1).

Q10 (Future Work): How would you implement closed-loop firing at a moving target? What are th

Likely Answer Grade: Pass

Model Answer:

The implementation would proceed through Stages S5/S6 of the integration protocol with the subject walking/jogging during the shot (Sec 6.4.1, p.60). The Kalman filter's predict-ahead output (already shipped in the UDP target packet at 200-400ms horizon) would aim at the predicted joint position at the moment the ball arrives, rather than the current position (Sec 6.4.2, p.61). Technical barriers include: (1) empirical RPM-to-velocity calibration is needed to know the flight time for the prediction horizon (Sec 5.9, p.55; Sec 6.4.3, p.61); (2) the CV Kalman model is neutral on jump sequences, requiring an IMM or ballistic motion model (Sec 5.7, p.53; Sec 6.4); (3) safety gating must be extended to reject shots when the target's velocity exceeds a safe threshold; (4) the 19/81 invalid trial rate due to occlusion must be addressed for reliable moving-target tracking (Sec 5.4).

Q11 (Methodology): Why use YOLO-Pose instead of MMPose? What was the ablation result?

Likely Answer Grade: Pass

Model Answer:

YOLO-Pose deployed via TensorRT FP16 produces a 6.2x live-inference speed-up over the MMPose two-stage pipeline: approximately 6.2ms per four-camera batch versus approximately 38.5ms per image for MMPose (Sec 5.6, p.52). This speed is critical for fitting both the ball detector and the pose estimator within the 15 FPS budget. The ablation showed that 3D jitter difference between the two backends remained within 5mm across walk, jog, and jump sequences, and per-camera 2D detection rates were within 6 percentage points (Sec 5.6). The accuracy degradation was judged acceptable for the latency gain. MMPose is retained as the reference backend for the joint-touch ground-truth analysis (Sec 4.2.3, p.39).

Q12 (Future Work): What is the RPM-to-velocity calibration protocol, and why is it urgent?

Likely Answer Grade: Pass

Model Answer:

The protocol (Sec 6.4.3, p.61) proposes two options: (A) Doppler radar gun for direct ball exit speed measurement per RPM setting, or (B) high-speed camera at 240+ fps for pixel-tracking over a known distance. The plan is 50-100 shots across the RPM range (0, 500, 1000, 1500, 2000 RPM per motor), fitting a polynomial or lookup table mapping commanded RPM to measured exit speed. It is urgent because: (1) the current ballistic solver assumes a fixed 10 m/s muzzle speed, which is accurate only at 500-600 RPM; at 800 RPM the nose shots in the live test were off-target (Sec 5.9, p.55); (2) predictive aiming for moving targets requires knowing the flight time, which depends on the muzzle speed (Sec 6.4.1, p.60). The current 95.17mm accuracy was achieved without this calibration, so empirical calibration is expected to improve results further (Sec 6.4.3). Safety RPM clamps remain regardless of calibration state (Sec 3.14.1).

SECTION 3: FINAL COMMITTEE SCORING SHEET

Presentation Score (50 points max)

Criterion | S1(35%) | S2(35%) | EE(30%) | Weighted Avg

1. Speech & Style (10) | 8.0 | 7.8 | 7.5 | 7.78
2. Structure (10) | 8.4 | 8.4 | 8.0 | 8.28
3. Visual Aids (10) | 7.6 | 7.5 | 7.0 | 7.38
4. Q&A (20) | 7.5 | 7.5 | 7.0 | 7.35
PRESENTATION SUBTOTAL | 31.5/50 | 31.2/50 | 29.5/50 | 30.79/50 = 61.6%

Technical Content Score (50 points max)

Criterion | S1(35%) | S2(35%) | EE(30%) | Weighted Avg

5. Introduction (10) | 7.5 | 7.5 | 7.0 | 7.35
6. Tech Competency(30)| 8.0 | 8.0 | 7.5 | 7.85
7. Concl/Future/Eth(10)| 7.0 | 7.0 | 6.5 | 6.85
TECHNICAL SUBTOTAL | 22.5/50 | 22.5/50 | 21.0/50 | 22.05/50 = 44.1%

Overall Oral Defense Score (O)

Presentation: 30.79/50 = 61.6%
Technical: 22.05/50 = 44.1%
ORAL TOTAL (O) = 52.84/100 = 52.84%

Scoring Notes (justification for deductions):

- Speech & Style: professional delivery but occasionally imprecise ('closed-loop' when aim-only validated)
- Structure: strong logical flow; docked for missing literature citations on motivation slide
- Visual Aids: several slides cluttered (Slide 2, 5, 8); good use of tables and pipeline diagrams
- Q&A: methodology answers strong; results answers would be weak on bias correction details; ethics shallow
- Introduction: problem statement clear but literature coverage thin on slides (relegated to spoken)
- Tech Competency: strong pipeline justification; cost analysis present; docked for 'closed-loop' overstatement
- Conclusions/Future/Ethics: dual-use statement missing from slide; standards listed without implementation detail

Final Evaluation Calculation

$$FE = 0.7 * M + 0.3 * O$$

Where O (oral) = 52.84%

Assuming M (manuscript) = 75% (conservative estimate based on thesis quality)

$$FE = 0.7 * 75 + 0.3 * 52.84 = 52.5 + 15.85 = 68.35\%$$

FE = 68.35% < 75% -> WOULD BE UNSUCCESSFUL if oral alone determined outcome

HOWEVER: The oral score above is a SIMULATION assuming an average defense.

With strong Q&A performance (8/10 on Q&A instead of 7.35):

$$\text{Revised O} = \sim 58\% \rightarrow FE = 0.7 * 75 + 0.3 * 58 = 52.5 + 17.4 = 69.9\% \text{ (still < 75\%)}$$

$$\text{Need O} \geq 83.3\% \text{ for FE=75\% if M=75\%: } 0.7 * 75 + 0.3 * 83.3 = 52.5 + 25 = 77.5\%$$

This means the oral defense must be STRONG to compensate.

TOP 5 RISKS THAT COULD DROP YOU BELOW 75%

RISK 1: Claiming 'closed-loop' or 'full autonomy' when the thesis explicitly states closed-loop firing at a moving subject is NOT validated (Sec 1.3, p.4; Sec 5.11, p.56). The EE will penalise this heavily under 'Viability of design' (Tech Competency 30p).

RISK 2: Ignoring or downplaying the $z = -103\text{mm}$ Z-axis bias and the in-sample correction. If you present 95.17mm without acknowledging the raw 150.77mm and the in-sample fitting limitation (Sec 4.4.2, p.41; Sec 6.3), the committee will question your understanding of the results under 'Interpretation of results'.

RISK 3: No cost discussion or vague cost figures. Your slide says ~\$200 but total BLM hardware is ~\$358. The handbook requires 'Costs and efficiency considerations' under Technical Competency (30p). Inconsistency here undermines credibility.

RISK 4: Missing safety interlocks detail on the ethics slide. Listing ISO numbers without explaining HOW they are implemented (NC E-STOP = Cat-1 stop, fuse sizing per IEC 60204-1, exclusion zone per ISO 10218) fails the handbook criterion 'Compliance with good practices and standards' (Conclusions 10p).

RISK 5: Omitting the dual-use statement. Sec 6.5 (p.64) is explicit about the weaponisation potential. Not addressing this in the defense will be seen as lacking professional ethics awareness under 'Compliance with good practices and standards' and the handbook's 'Professional ethics' criterion.

VERDICT

ORAL DEFENSE (simulated, average Q&A): UNSUCCESSFUL (52.84/100)

ORAL DEFENSE (with strong Q&A preparation): POTENTIALLY SUCCESSFUL

CRITICAL ACTIONS BEFORE DEFENSE:

1. Remove ALL 'closed-loop' claims from slides; replace with 'aim-only + controlled single-shot'
2. Add dual-use statement to ethics slide (quote Sec 6.5)
3. Show raw vs corrected numbers (150.77 -> 95.17mm) with in-sample caveat
4. Clarify cost: perception \$120 vs total \$358
5. Add implementation details to standards: NC E-STOP = ISO 13849 Cat-1, etc.
6. Practice Q&A with emphasis on bias correction, safety layers, and dual-use

With these corrections, the oral score could reach 70-75/100,

yielding $FE = 0.7 \cdot 75 + 0.3 \cdot 72 = 52.5 + 21.6 = 74.1\%$ (borderline)

or with $M=80\%$: $FE = 0.7 \cdot 80 + 0.3 \cdot 72 = 56 + 21.6 = 77.6\%$ (SUCCESSFUL)

The manuscript quality (M) is likely the deciding factor.

A well-prepared oral defense with honest framing of limitations can pass.